



ABT RECERTIFICATION EXAMINATION

May 10, 2013

PART A

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 2004 and 2009. There are three parts to the exam. An answer sheet will be mailed for each part. These answer sheets must be returned to ABT, P.O. Box 97786, Raleigh, NC 27624 for grading by **July 26, 2013**. If you wish to return them by courier, the physical address is 2806 Treasures Lane, Raleigh, NC 27614. If you do not achieve a passing score on any of the parts, you will be contacted and given another opportunity to pass. If you pass all three parts, your scores will be provided to you in November of this year, along with the results of the Recertification Committee's review of your other application materials. If you have any questions please contact the ABT office at (919) 841-5022 or info@abtox.org.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.
2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.
3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology, molecular toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet **ONLY** on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

Good Luck!

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*Key for the following questions
Choose the single most appropriate answer*

1. Measurement of which of the following in hair can be used as a reliable indicator of exposure?
 - A) selenium
 - B) mercury
 - C) lead
 - D) cadmium
 - E) copper
2. Which of the following metals has NOT been shown to be carcinogenic in humans?
 - A) arsenic
 - B) mercury
 - C) chromium
 - D) nickel
 - E) cadmium
3. Which of the following BEST describes berylliosis (Chronic Beryllium Disease - CBD)?
 - A) chest radiographic abnormalities are a sensitive measure of early CBD
 - B) immunological sensitization is associated with pulmonary effects
 - C) the characteristic lesion is small-cell carcinoma
 - D) experimental studies indicate a link to characteristic mutations in K-ras
 - E) berylliosis is a classic dose-response disease
4. Which of the following is the most common adverse health effect from nickel exposure?
 - A) contact dermatitis
 - B) peripheral neuropathy
 - C) cancer
 - D) fatty liver
 - E) hypertension
5. Which of the following is the primary mechanism of action for mercury-induced non-specific cell injury or cell death?
 - A) high-affinity binding of divalent mercury to sulfhydryl groups of proteins
 - B) uncoupling of mitochondrial oxidative phosphorylation
 - C) cellular mimicry and replacement of divalent zinc in essential zinc metalloenzymes
 - D) formation of DNA adducts or DNA-protein crosslinks
 - E) membrane lipid peroxidation
6. Which of the following is NOT CORRECT regarding solvents?
 - A) intentional exposure to inhaled solvents (abuse) can cause visual as well as auditory hallucinations
 - B) benzene and toluene are the two most commonly identified volatile organic compounds in indoor air
 - C) the finding of a great many solvents in drinking water derived from the Mississippi River prompted the passage of the Safe Drinking Water Act in 1974
 - D) the majority of solvent release into the atmosphere occurs during misuse and accidental release
 - E) nearly all hydrocarbon solvents can cause CNS depression
7. Which of the following signs is/are associated with industrial exposure to vanadium pentoxide dust?
 - A) encephalopathy and cerebral edema
 - B) burning feet syndrome and paresthesia
 - C) painful osteomalacia and osteoporosis
 - D) greenish-black discoloration of the tongue
 - E) pink coloring of the hands and feet
8. Which of the following is NOT CORRECT regarding methyl tertiary butyl ether (MTBE)?
 - A) MTBE is well absorbed following oral, dermal and inhalation exposures in humans
 - B) carcinogenicity studies in rats and mice by inhalation and oral routes showed no tumor formation
 - C) oxidation of MTBE in humans is primarily by CYP2A6
 - D) the two major urinary metabolites of MTBE are 2-methyl-1, 2 propanediol and 2-hydroxyisobutyrate
 - E) although formaldehyde is one of the oxidative metabolites, it is undetectable in humans and rats following exposure to MTBE
9. Which of the following is NOT CORRECT regarding arsenic poisoning?
 - A) in general, the trivalent form of arsenic is less toxic than the pentavalent form
 - B) acute inorganic arsenic toxicity includes vasodilation
 - C) acute inorganic arsenic poisoning induces diarrhea
 - D) chronic arsenic ingestion produces hyperpigmentation of the skin
 - E) the most common arsenic-induced neurological lesion is peripheral neuropathy

10. Which of the following is consistent with the generally accepted mode of action of carcinogenicity for trichloroethylene (TCE) exposure in animals?

- A) the metabolite trichloroethanol glucuronide (TCOH-glucuronide) has been shown to cause uterine tumors in rats following oral exposure
- B) the metabolite 1,2-dichlorovinyl-L-cystine (DCVC) has been shown to cause renal tumors in female mice following inhalation exposure
- C) the metabolites dichloroacetate (DCA) and trichloroacetate (TCA) have been shown to cause liver tumors in Sprague-Dawley rats following inhalation exposure
- D) the metabolite chloral has been shown to cause lung tumors in mice following inhalation exposure
- E) regardless of the intermediate metabolites, animal models show the same hepatic and renal tumors reported in occupational cohorts exposed to TCE

11. Exogenous factors can influence the toxicity of solvents. Which of the following is NOT CORRECT?

- A) dietary lipids in the GI lumen can diminish the toxicity of carbon tetrachloride by delaying absorption
- B) increased alveolar ventilation during exercise increases absorption of polar solvents because of their high blood:air partition coefficients
- C) endotoxin potentiates the liver injury caused by ethanol and other solvents by triggering the release of inflammatory mediators which may make cells more susceptible to damage
- D) pre-treatment of rats with 2-butanol potentiates the toxicity of carbon tetrachloride through P450 induction
- E) administration of chloroform at the beginning of a rat's activity cycle leads to decreased liver injury because hepatic GSH levels are lower at the beginning of the activity cycle

12. Which of the following is CORRECT regarding insect-resistant corn?

- A) *Bacillus thuringiensis* is incorporated into the corn
- B) a protein derived from *Bacillus thuringiensis* is sprayed on corn
- C) currently much of the corn in the USA makes the insecticidal protein
- D) the use of *Bacillus thuringiensis* on the nutritional value of corn is not known
- E) the precursors that make the insecticidal protein for corn are considered toxins

13. Which of the following is the BEST approach to limiting solvent overexposure in an occupational setting?

- A) local exhaust ventilation
- B) personal protective equipment
- C) dilution ventilation
- D) process change/solvent elimination
- E) frequent job rotation

14. Which of the following BEST describes the reason for pesticide poisoning in developed countries?

- A) they are deliberately added to the environment for the purpose of killing some form of life
- B) many pesticides are too toxic to be used safely
- C) pesticides are improperly used and stored in many households
- D) pesticides exist in formulations that contain multiple ingredients
- E) increased use of pesticides from year to year

15. Which of the following is the MOST effective treatment for mammalian organophosphate (OP) toxicity?

- A) 2-PAM
- B) diazepam
- C) epinephrine
- D) atropine
- E) curare

16. Glutamine synthetase is a key enzyme found in plant and animal systems and functions in the utilization of nitrogen inside cells. The herbicide glufosinate inhibits this enzyme resulting in which of the following?

- A) a toxic build up of ammonia in humans but not plants
- B) a toxic build up of ammonia in plants but not humans
- C) a toxic build up of ammonia in both plants and humans
- D) a toxic build up of glutamine in plants but not humans
- E) no impact as alternative pathways exist in both plants and humans to compensate for inhibition of glutamine synthetase

17. Which of the following is NOT CORRECT regarding the toxicity of manganese?

- A) it causes cirrhosis of the liver
- B) less than 5% of ingested manganese is absorbed, but the absorption increases with iron deficiency
- C) the squirrel monkey provides a model of the central nervous system syndrome resulting from inhalation
- D) it is a recognized human carcinogen
- E) occupational exposure to it is associated with a 30-fold elevation in respiratory disease

18. The toxic signs and symptoms of pyrethroids are associated with which of the following?
- A) oxidative metabolism to an active form
 - B) interference of the normal hormones
 - C) activation of a α_2 -adrenergic receptors
 - D) disruption of voltage-gated sodium channels
 - E) inhibition of acetylcholinesterase (AChE)
19. Terrestrial environmental exposure to organic tin compounds is most likely to be associated with the use of which of the following?
- A) fungicides
 - B) paint
 - C) feed additives
 - D) machinery grease
 - E) fumigants
20. Which of the following enzymes is involved in organophosphate-induced delayed polyneuropathy?
- A) acetylcholinesterase
 - B) butyrylcholinesterase
 - C) carboxylesterase
 - D) neuropathy target esterase
 - E) ATPase
21. Inhalation exposure to SO_2 gas may result in which of the following?
- A) bronchoconstriction via Type I hypersensitivity
 - B) decreased mucociliary clearance
 - C) local and centrally mediated bronchoconstriction via formation of sulfite
 - D) autooxidation of cell membranes via free-radical formation
 - E) induction of pulmonary fibrosis following damage to epithelial cells
22. Inhaled aerosols and particulates may be absorbed in various regions of the respiratory tract. Which two important characteristics affect absorption of aerosols after exposure?
- A) aerosol size and water solubility
 - B) tissue reactivity and water solubility
 - C) aerosol size and tissue reactivity
 - D) water solubility and pH of the aerosol
 - E) pH of the aerosol and aerosol size
23. Which of the following substances is an HDL-raising agent that inhibits the lipolysis of triglycerides by hormone-sensitive lipase, thereby reducing transport of free fatty acids to the liver and decreasing hepatic triglyceride synthesis?
- A) cholestyramine
 - B) niacin
 - C) clofibrate
 - D) avasimibe
 - E) lovastatin
24. Which of the following pertaining to warfarin and related coumarin derivatives is CORRECT?
- A) there are no differences in half-lives or potency of the two enantiomers (-R and S-enantiomers)
 - B) genetic polymorphisms in CYP2C9 may affect the metabolism of the S-enantiomer and have functional consequences as it pertains to bleeding
 - C) APTT (activated partial thromboplastin time) is measured to evaluate the adequacy of the dosage of warfarin
 - D) xenobiotics that result in displacement of warfarin from albumin would be expected to lessen the tendency for bleeding
 - E) there are no teratologic risks associated with warfarin administration
25. For which agent listed below is the neurotoxicity associated with its use mediated by a disruption of microtubule dissociation into tubulin subunits?
- A) atropine
 - B) d-tubocurarine
 - C) lysergic acid diethylamide
 - D) paclitaxel
 - E) procainamide
26. Ionization frequently disrupts chemical bonding in cellular molecules such as DNA. Which of the following is the most common type of DNA damage caused by gamma (low-linear energy transfer) radiation exposure in mammalian cells?
- A) double-strand breaks
 - B) base damage
 - C) single-strand breaks
 - D) DNA protein cross-links
 - E) pyrimidine dimers
27. Soft tissue calcification in cattle has been attributed to which of the following?
- A) eating plants contaminated by polychlorobiphenyls
 - B) eating plants containing Vitamin D-like glycosides
 - C) eating feed supplemented with high levels of calcium
 - D) eating feed contaminated by aflatoxin
 - E) drinking water contaminated by glyphosate
28. Which of the following properties best describes the estrogenic activity of phytoestrogens?
- A) antigonadotropic effects at high doses
 - B) binding to intracellular receptors at the same affinity as estradiol
 - C) causing long-term reproductive effects in humans
 - D) becoming more potent than estradiol due to activation by P-450 enzymes
 - E) affecting only female gonadal tissue in animals

29. The particle size of an aerosol is a major determinant of where in the respiratory tract the aerosol is likely to deposit and cause a possible toxic response to the lung. The particle size of an aerosol is usually measured by determining the CMD and MMAD. In addition, the σ_g is determined as a measure of the range of distribution of particles. Which of the following is NOT CORRECT regarding aerosol measurements and characteristics?

- A) the effect of mass and airflow is reflected in the MMAD measurement
- B) particles in the MMAD of 5 μm or greater usually deposit in the nasopharyngeal region
- C) an aerosol with a σ_g less than or equal to 1.2 is considered to be monodisperse
- D) from a log-probit plot, $\sigma_g = 84.1\%$ size/50%size
- E) an aerosol with a $\sigma_g = 2.0$ is considered to be monodisperse

30. Snake venoms are complex mixtures, however, their poisonous effects are most commonly due to which of the following substances?

- A) enzymes and polypeptides
- B) formic acid
- C) tetrodotoxin
- D) toxic alkaloids
- E) vasoactive amines

31. Food idiosyncrasy is characterized by which of the following?

- A) quantitatively normal response to a food substance or contaminant
- B) an immunologic mechanism
- C) more likely to occur in genetically predisposed individuals
- D) a reaction qualitatively similar to the normal physiologic response to a substance
- E) a humoral immune response involving IgE

32. Bisphenol A is considered a chemical of concern due to which of the following?

- A) use in plastics and strong binding to the aryl hydrocarbon receptor
- B) strong binding to the estrogen receptor and induction of biotransformation enzymes
- C) extensive use in plastic manufacturing and weak binding to the estrogen receptor
- D) surfactant properties and potential for endocrine disruption
- E) bioaccumulation in the environment and potential as a chemical carcinogen

33. Which of the following statements about selenium and selenium compounds is NOT CORRECT?

- A) selenium's actions in antioxidant systems make it an essential trace element
- B) acute exposure to selenium causes blind staggers in cattle
- C) selenium enhances the toxic effects of methylmercury
- D) chronic exposure to excess selenium can lead to teeth, hair, and nail loss
- E) accumulated evidence indicates that selenium compounds possess anticancer properties

34. Which of the following is an important class of chemicals that are formed during grilling or broiling of meat and is associated with cancer in humans?

- A) acrylamide
- B) heterocyclic amines
- C) aliphatic amines
- D) ethyleneimine
- E) hydrazines

35. Which aflatoxin is the most toxicologically potent and commonly occurs in produce?

- A) Aflatoxin B₁
- B) Aflatoxin B₂
- C) Aflatoxin C₁
- D) Aflatoxin G₁
- E) Aflatoxin G₂

36. Rhabdomyolysis can be the direct result of a toxicant or the indirect result of a drug overdose-induced muscular ischemia. Please select the statement that is NOT CORRECT with regard to rhabdomyolysis.

- A) hyperkalemia and hyperphosphatemia are conditions of being predisposed to developing rhabdomyolysis
- B) patients with rhabdomyolysis will first exhibit a rise in serum myoglobin and then an increase in creatine kinase
- C) patients with rhabdomyolysis who have myoglobin urine concentrations can have red colored urine
- D) patients with high levels of cocaine use may develop rhabdomyolysis
- E) rhabdomyolysis may result from severe carbon monoxide poisoning

37. Which of the following is a specific antidote for benzodiazepine overdose?

- A) buprenorphine
- B) rifampin
- C) flumazenil
- D) dextroamphetamine
- E) carbamazepine

38. Some ethnic populations, such as Asians and Native Americans, have difficulty metabolizing alcohol. The majority of the affected individuals have a defect in which liver enzyme?

- A) CYP1A2
- B) CYP2E1
- C) catalase
- D) alcohol dehydrogenase
- E) acetaldehyde dehydrogenase

39. Which of the following is CORRECT regarding the mycotoxin food contaminant zearalenone?

- A) it is mutagenic
- B) it is produced by *Aspergillus flavus*
- C) it decreases the reproductive potential in farm animals
- D) it causes leukoencephalomalacia in horses
- E) it requires metabolic activation to cause adverse effects

40. Which of the following toxicities is NOT associated with acute or chronic tetracycline antibiotics use?

- A) blood dyscrasias
- B) skin and mucosal reactions
- C) cardio-pulmonary effects
- D) liver effects
- E) kidney effects



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PART B

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Choose the single most appropriate answer

1. Which of the following genetic toxicology assays measures both mutagenic and clastogenic endpoints?
 - A) Ames test
 - B) Mouse Micronucleus test
 - C) Mouse Lymphoma assay
 - D) Comet assay
 - E) Sister Chromatid Exchange assay
2. Which of the following is NOT CORRECT regarding the proposed mode of action for the development of hemangiosarcoma induced by nongenotoxic compounds?
 - A) local tissue hypoxia
 - B) lymphocyte activation and release of chemokines
 - C) dysregulated angiogenesis/hematopoiesis
 - D) increased circulating endothelial cells
 - E) enhanced endothelial cell proliferation
3. Which of the following is NOT CORRECT regarding the mouse lymphoma L5178Y cell assay for mutagenesis in the TK gene?
 - A) the TK gene allows cells to scavenge thymidine for use in DNA synthesis
 - B) the assay selects for the TK^{+/+} phenotype
 - C) the assay selects for forward mutations in the TK gene
 - D) the assay is part of the genotoxicity battery proposed by the International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use, and has been accepted by the US FDA
 - E) the TK gene metabolizes 5-trifluorothymidine to a toxin, resulting in cell death
4. Which of the following is considered evidence of a mutagenic event and not a type of DNA damage?
 - A) adduct formation
 - B) intercalation
 - C) base substitution
 - D) cross-links
 - E) strand breaks
5. Which of the following is CORRECT regarding DNA repair in mammalian cells?
 - A) it is identically efficient in all species and tissues
 - B) it is more efficient in genes being transcribed
 - C) it repairs all types of DNA damage with equal efficiency
 - D) it corrects an existing mutation in the DNA molecule
 - E) it is not inducible by exogenous agents
6. Which one of the agents listed below results in steatosis?
 - A) lead
 - B) valproic acid
 - C) vitamin K
 - D) vancomycin
 - E) chloramphenicol
7. Which of the following statements is NOT correct?
 - A) cDNA microarrays use RNA samples to compare gene expression levels between two samples
 - B) 2D gel electrophoresis and mass spectrometry are the two major analytical tools used in proteomic studies
 - C) microRNA arrays identify low-molecular weight RNAs involved in mRNA degradation, transcriptional gene silencing and translational repression
 - D) metabonomic studies have established well-defined biomarkers of exposure useful in animals and epidemiological studies
 - E) antibody arrays measure protein expression similar to that of western blots
8. What is the purpose of including cytochalasin B in the *in vitro* micronucleus assay?
 - A) to improve assay sensitivity by increasing cellular proliferation rates
 - B) to block cytokinesis and avoid confusion associated with differences in cell proliferation kinetics
 - C) to facilitate automated scoring by improving uptake of stains for micronuclei in the treated cells
 - D) to decrease scoring errors by minimizing the incidence of binucleate and multinucleate cells
 - E) to detect kinetochores and enable distinguishing between micronucleus formation by aneugenic versus clastogenic mechanisms
9. When assessing peri- and post-natal developmental toxicity using ICH guidelines, which of the following is NOT normally measured?
 - A) duration of pregnancy
 - B) live offspring per litter
 - C) number of implantations
 - D) body weights at birth
 - E) corpora lutea counts
10. In an embryo-fetal development study, during which of the following periods is it important to expose the pregnant females to the test agent?
 - A) throughout the entire gestation period
 - B) during the implantation period
 - C) pre and post fertilization of the female
 - D) during organogenesis
 - E) during mating
11. The majority of adverse reproductive outcomes have been traced to which of the following?

- A) chemical and other environmental exposures
 - B) the significant use of pharmaceuticals by pregnant women in the first trimester
 - C) maternal infections
 - D) unknown etiologies
 - E) alcohol
12. Which of the following BEST describes endocrine disrupters?
- A) are a very new concern in toxicology
 - B) generally act by receptor competition or inhibition of steroidogenesis
 - C) only interfere with female reproduction
 - D) lead to increased sperm counts in humans
 - E) do not affect reproductive tract organogenesis
13. Drugs and chemicals can produce a variety of effects on the immune system, including allergic skin sensitization, production of autoimmune disease, and immunosuppression. Which of the statements below is an example of an effect which is consistent with immunosuppression caused by a drug or chemical?
- A) development of hemolytic anemia in response to drug therapy with penicillin
 - B) increased T-cell proliferation by known skin sensitizers in the Murine Local Lymph Node Assay
 - C) IgM antibody production and plaque formation after immunization with sheep red blood cells in a Plaque Antibody Forming assay
 - D) histological evidence of hypocellularity in the spleen, thymus and lymph nodes in a toxicity study
 - E) increased production of IgM antibody in the systemic circulation, after immunization with sheep red blood cells
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15. Which of the following BEST characterizes the findings of toxic response to *in utero* exposure to ethanol?
- A) the occurrence of a few specific, severe developmental abnormalities
 - B) mechanisms of toxicity that have been experimentally well characterized and linked to oxidation of inositol diacetylphosphate
 - C) fetal alcohol syndrome, a toxicity also seen with exposure to other substances
 - D) dose-related reduction in fetal body weights
 - E) effects observed only at high exposure levels
16. Along the upper and lower airways, which of the following cell types are the most sensitive to ozone?
- A) ciliated cells and Clara cells
 - B) ciliated cells and Type 1 cells
 - C) ciliated cells and Type 2 cells
 - D) Clara cells and Type 1 cells
 - E) Clara cells and Type 2 cells
17. Which of the following compounds/agents is a direct activating carcinogen that does not require metabolic activation to induce cancer?
- A) azo dyes
 - B) nitrogen mustard
 - C) aflatoxin
 - D) aromatic amines
 - E) n-nitrosamines

18. Naphthalene induces extensive necrosis in the bronchiolar epithelium in mice, but much less in rats and hamsters. This response is believed to be due to which of the following?
- A) anatomical difference that result in a higher tissue dose in the mouse
 - B) CYP2F2, a unique isoform, rapidly activates naphthalene in mice
 - C) a unique aryl hydrocarbon receptor in the mouse
 - D) absence of glutathione transferase in mouse bronchiolar epithelium
 - E) rapid oxidation by myeloperoxidase in mice
19. Which of the following is NOT CORRECT regarding the blood brain barrier?
- A) lipid solubility affects penetration of compounds into the brain
 - B) xenobiotic transporters contribute to the blood brain barrier
 - C) endothelial tight junctions are a critical component
 - D) the circumventricular organs are not protected by the blood brain barrier
 - E) unconjugated bilirubin crosses the blood brain barrier in adults more readily than in children
20. The mechanism by which ozone causes *acute* cellular injury in the respiratory tract is best described by which of the following?
- A) hyperplasia and hypertrophy of type I alveolar cells and alterations of Clara cells in the small airways
 - B) inhibition of macrophage function and mucociliary clearance
 - C) oxidation of fatty acids in cell membranes leading to lipid radicals and autooxidation of cell macromolecules
 - D) endothelial dysfunction leading to cardiac effects including changes in heart rate and heart rate variability
 - E) inhibition of oxidative phosphorylation
21. Which of the following produces amnesic (memory-loss) shellfish poisoning?
- A) domoic acid
 - B) ciguatera toxin
 - C) bovine spongiform encephalopathy prion
 - D) *Clostridium botulinum*
 - E) tetrodotoxin
22. Phagocytic cells that are part of the innate immune system of mammals include which of the following?
- A) B cells
 - B) natural killer (NK) cells
 - C) polymorphonuclear cells (neutrophils)
 - D) follicular dendritic cells (FDCs)
 - E) T cells
23. Which of the following characteristics is common to all four types of hypersensitivity reactions?
- A) the generation of T cells
 - B) production of IgE
 - C) production of IgG
 - D) prior exposure to sensitizers
 - E) the generation of B cells
24. Analgesics such as ibuprofen, aspirin, and cyclooxygenase-2 inhibitors exhibit both acute and chronic renal toxicity. Analgesic nephropathy is often irreversible and is best characterized by necrotic lesions in which of the following?
- A) the glomerulus
 - B) the proximal tubule
 - C) the distal tubule
 - D) the renal papilla
 - E) the renal arteriole
25. Which of the following best describes the characteristics of immunoglobulin-G (IgG) found in human serum?
- A) the first antibody produced in an immune response to an infectious agent or vaccine
 - B) predominantly synthesized during a secondary immune response to an infectious agent or vaccine and can cross the placenta to provide protective immunity to the newborn infant
 - C) the primary antibody mediator of allergic reactions and atopic disease in humans
 - D) produced by polymorphonuclear neutrophils (PMNs) and macrophages at the site of an infection
 - E) an increase in total IgG levels can be an indication of immunosuppression

26. Which of the following is CORRECT about the immune system, immunoresponsiveness, and the evaluation of immunotoxicity in minipigs?

- A) when compared with dogs and nonhuman primates, the greatly reduced interanimal variability in functional immune response (e.g., TDAR) makes the minipig a more desirable non-rodent model to assess immunotoxicity
- B) the immune system of the minipig has been demonstrated to be sufficiently different from humans so as to be an inappropriate model for vaccine and vaccine adjuvant development
- C) in validation studies similar to those used to validate the rat as a model for general immunotoxicity testing, quantitative methods to assess functional responses were investigated successfully; however, immunophenotyping has been most successful since most antibodies that cross-react in nonhuman primates also cross-react in minipigs
- D) the immune system has generally been well characterized from an anatomic and clinical pathology perspective as well as through a comparative assessment of immune-response genes in the mouse, minipig, and human
- E) the stress response in the minipig involves the lymphocytes but does not appear to result in changes in the monocytes, neutrophils, or the thymus

27. The primary factors that determine the rate of absorption of a substance applied to the surface of the skin of a test animal include which of the following?

- A) age of test animal, weight of the test animal, concentration of the dosing solution
- B) age of test animal, weight of the test animal, anatomical site of application
- C) sex of test animal, anatomical site of application, pH of the dosing solution
- D) anatomical site of application, concentration of the dosing solution, pH of the skin
- E) age of test animal, anatomical site of application, concentration of the dosing solution

28. Dermal hyperpigmentation is BEST attributed to which of the following agents?

- A) furocoumarins (psoralens)
- B) butylated hydroxytoluene
- C) TCDD (2,3,7,8-tetrachlorodibenzo-p-dioxin)
- D) mercaptoamines
- E) p-tertbutylphenol

29. Which of the following is CORRECT regarding eye irritation studies in rabbits?

- A) can detect only irreversible changes
- B) produces high interlaboratory scoring consistency
- C) correlate most closely with human results than studies in any other species
- D) may be conducted by placing the test substance in the conjunctival sac
- E) are completed after a 24-hour observation

30. The murine local lymph node assay is BEST described by which of the following?

- A) can determine the elicitation threshold of a potential dermal sensitizer
- B) is a confirmatory test to the guinea pig maximization test or Buehler test
- C) requires an adjuvant to boost the immune system and increase sensitivity
- D) is a validated method for measuring photosensitization potential
- E) measure lymphocyte proliferation as expressed by the stimulation index

31. Liver toxicants can cause injury via numerous mechanisms. Considering fatty liver (steatosis) as a manifestation of liver injury, which phrase best describes the progression of injury beyond the initial stage?

- A) steatosis -> steatohepatitis -> fibrosis -> cirrhosis
- B) steatosis -> fibrosis -> steatohepatitis -> cirrhosis
- C) steatosis -> steatohepatitis -> cirrhosis -> fibrosis
- D) steatosis -> cirrhosis -> steatohepatitis -> fibrosis
- E) steatosis -> necrosis -> fibrosis -> cirrhosis

32. Which of the following agents cause acute renal failure due to deposition of calcium monohydrate crystals in kidney tubules?

- A) ethylene glycol
- B) cadmium
- C) ochratoxin
- D) d-limonene
- E) cisplatin

33. An anti-neoplastic agent used to kill proliferating cancer cells also kills proliferating bone marrow precursor cells of all 3 major cell lineages (erythroid, myeloid, and megakaryotytic) as an undesired side effect. The agent is equally cytolethal to precursors of all 3 lineages, and is rapidly cleared from the body, allowing repopulation of precursor cells from marrow stem cells. Assuming no additional effects of the agent and equal rates of repopulation of the various marrow precursor cells, what circulating cell population in the peripheral blood would be expected to take the longest to reach its nadir (low point) following a single treatment with the agent?
- A) neutrophils
 - B) eosinophils
 - C) erythrocytes
 - D) monocytes
 - E) platelets
34. The mechanism of hepatotoxicity related to ingestion of excessive amounts of Vitamin A is due to which of the following?
- A) accumulation of lipid within parenchymal hepatocytes and subsequent mitochondrial toxicity
 - B) stellate cell hypertrophy leading to sinusoidal occlusion
 - C) activation of stellate cells resulting in hepatocyte death and fibrosis
 - D) formation of reaction oxygen species in Kupffer cells and release of cytotoxic cytokines
 - E) depletion of retinol binding protein allowing free retinoic acid to react with biliary transport proteins leading bile duct injury
35. Chemicals such as chloramphenicol, phenylbutazone, and phenytoin can cause a condition known as aplastic anemia. The cellular target of these chemicals resulting in this condition is which of the following?
- A) hematopoietic stem cell (CFU-S)
 - B) colony forming unit-granulocyte (CFU-G)
 - C) colony forming unit-erythrocyte (CFU-E)
 - D) polymorphonuclear cells (PMN)
 - E) promyelocyte
36. Both the US FDA and the European Medicines Agency (EMA) approved the use of seven novel urinary biomarkers as valid predictors of nephrotoxic injury in animal studies. Which of the following is NOT one of the recently approved biomarkers?
- A) total protein
 - B) clusterin
 - C) kidney injury molecule-1 (KIM-1)
 - D) creatine kinase
 - E) trefoil factor 3
37. Amphotericin B nephrotoxicity can best be described by which of the following?
- A) papillary necrosis, interstitial nephritis, oliguria
 - B) glomerular and tubular injury, arterial vasodilation, ADH-resistant polyuria
 - C) proximal tubule S1/S2 necrosis, renal phospholipidosis, polyuria
 - D) proximal tubule S3 necrosis, renal magnesium wasting, polyuria
 - E) proximal tubule necrosis, decreased paraaminohippurate clearance, increased Na/K excretion
38. Along with acute and chronic renal failure, renal magnesium wasting and polyuria are typical of toxicity from which of the following?
- A) chloroform
 - B) acetaminophen
 - C) aminoglycosides
 - D) cisplatin
 - E) mercuric chloride (HgCl₂)
39. Which of the following is a mechanism associated with development of renal tubular lesions common to both chloroform and tetrafluoroethylene?
- A) bioactivation and covalent binding of reactive metabolites to cellular macromolecules
 - B) biotransformation by cysteine conjugate B-lyase
 - C) metabolism by cytochrome P450
 - D) proximal tubular reabsorption
 - E) alpha-2u-globulin accumulation in the proximal tubule
40. Angiotensin converting enzyme (ACE) inhibitors are administered for persons with heart failure. They can cause acute renal failure in patients with renal artery stenosis through which of the following mechanisms?
- A) ACE inhibitors block vasoconstriction, leading to decreased filtration pressure
 - B) increased production of red blood cells overwhelms glomerular filtration
 - C) decreased pressure associated with loss of vasoconstriction leads to increased glomerular filtration rate
 - D) vasoconstriction is increased with decreased prostaglandin synthesis, leading to increased renal blood flow
 - E) formation of hyaline droplets from ACE inhibitors binding to alpha-2u-globulin



ABT RECERTIFICATION EXAMINATION

May 10, 2013

PART C

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 2004 and 2009. There are three parts to the exam. An answer sheet will be mailed for each part. These answer sheets must be returned to ABT, P.O. Box 97786, Raleigh, NC 27624 for grading by **July 26, 2013**. If you wish to return them by courier, the physical address is 2806 Treasures Lane, Raleigh, NC 27614. If you do not achieve a passing score on any of the parts, you will be contacted and given another opportunity to pass. If you pass all three parts, your scores will be provided to you in November of this year, along with the results of the Recertification Committee's review of your other application materials. If you have any questions please contact the ABT office at (919) 841-5022 or info@abttox.org.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.
2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.
3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology, molecular toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet **ONLY** on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

Good Luck!

Key for the following questions
Choose the single most appropriate answer

1. Calculate the achieved dosage in mg/kg/day of a chemical fed to rats at a concentration of 0.5% (5000 ppm) in the diet. The rats had a mid period group mean body weight of 250g and ate 210g/week of the chemical diet admixture.

- A) 600
- B) 420
- C) 6000
- D) 4200
- E) 60

2. Select the correct pairing of toxic agent and mechanism of cardiotoxicity.

- A) toluene and sodium channel blockade
- B) carbon tetrachloride increased parasympathetic activity
- C) cocaine and calcium channel blockade
- D) cobalt altered calcium homeostasis
- E) barium and potassium channel blockade

3. Furanocoumarins in grapefruit juice can increase the systemic exposure to certain orally administered drugs such as simvastatin (HMG-coA reductase inhibitor) and felodipine (calcium channel blocker) through which of the following mechanisms?

- A) inhibition of intestinal CYP3A4
- B) inhibition of hepatic CYP3A4
- C) induction of UDP-glucuronosyltransferase
- D) inhibition of hepatic CYP1A1
- E) induction of aldehyde dehydrogenase

4. Many chemicals require metabolic activation to form their ultimate toxicants. Which of the following xenobiotics does NOT match its ultimate toxicant?

- A) fluoroacetate forms fluoronitrate
- B) acetaminophen forms N-acetyl-p-benzoquinoneimine
- C) carbon tetrachloride forms CCl_3OO
- D) benzo[a]pyrene forms benzo[a]pyrene-7-8-diol-9,10-epoxide
- E) ethylene glycol forms oxalic acid

5. Oligomycin, cyhexatin, DDT, and chlordecone all can interfere with mitochondrial ATP synthesis which can lead to ATP depletion and cell death. These agents impair mitochondrial ATP synthesis by which of the following mechanisms?

- A) they inhibit ADP phosphorylation by acting on ATP synthase
- B) they inhibit electron transport complex I (NADH-coenzyme Q reductase)
- C) they inhibit oxygen delivery to the electron transport chain
- D) they impair the transcription of key mitochondrial proteins
- E) they inhibit electron transport by acting on complex IV (cytochrome oxidase)

6. Establishing a causal association between an occupational toxicant and harmful effect/disease can be difficult. In order to evaluate the level of exposure at which there is an acceptable risk for health impairment, toxicologic data from a variety of sources is required, including in vitro assays, animal toxicology studies, human challenge studies, case reports and epidemiology studies. In animal studies, which one of the following is NOT a limitation?

- A) animal testing can provide only an estimate of the toxicity of a chemical that would be exhibited in humans
- B) animals do not always respond to a chemical exposure in the same way that humans would respond
- C) there can be interspecies differences in metabolism where cancer occurs in rodents but not humans
- D) relationships between the metabolic pathways of a chemical and its interaction with target molecules can be derived
- E) there are a few chemicals for which predictive animal models have either not yet been determined or do not exist

7. Which of the following would be expected to be the most effective treatment for protecting against systemic effects from hydrofluoric acid (HF) burns?

- A) sodium gluconate
- B) potassium gluconate
- C) calcium gluconate
- D) ethylenediaminetetraacetic acid (EDTA)
- E) sodium nitrite

8. Which of the following is NOT considered an important factor affecting nanoparticle toxicity in the lung?

- A) the large surface area relative to mass
- B) glucuronidation of nanoparticulates
- C) particulate diameters of less than 100nm
- D) the presence of reactive metals on the surface of nanoparticulates
- E) breathing rate

9. Which of the following is CORRECT regarding apoptosis or necrosis processes?

- A) ATP elevation initiates apoptosis
- B) ATP depletion is more likely to lead to necrosis
- C) apoptosis is dependent upon inhibition of RNA polymerase
- D) release of cytochrome C from the mitochondria is the initiating step of necrosis
- E) necrosis involves Mg^{++} dependent enzymatic processes

10. The drug thalidomide was originally marketed in Europe and Australia as a sedative for pregnant woman, but was banned in 1962 due to devastating birth defects (phocomelia). Which of the following is the primary mechanism of thalidomide teratogenicity?

- A) proliferation of hematopoietic stem cells
- B) inhibition of angiogenesis
- C) double strand breaks in proliferating cells
- D) cytotoxicity to progenitor osteoblasts
- E) stimulation of autoimmunity

11. Perchlorate is a goitrogen because it does which of the following?

- A) it blocks the action of thyroid peroxidase
- B) it blocks the transformation of inorganic iodide to molecular (reactive) iodine
- C) it competitively inhibits the absorption of iodine from the gut
- D) it competitively inhibits the action of the sodium-iodide symporter
- E) it decreases the release of thyroid stimulating hormone (TSH)

12. Transporters mediate the cellular influx or efflux of xenobiotics and other chemicals. One of the first active, energy-dependent xenobiotic transporters identified was P-glycoprotein (P-gp). Which of the following BEST describes this transporter?

- A) is an influx transporter that functions to regulate intracellular sodium concentration
- B) facilitates simple diffusion through the phospholipid bilayer of cell membranes
- C) mediates the influx of small hydrophilic molecules such as glycine
- D) is an efflux pump that can confer multidrug resistance to cancer cells
- E) increases the intestinal absorption of certain orally administered drugs

13. What is the most widely known factor that can play a major role in the differential response to isoniazid?

- A) polymorphism of the CYP2D6 isoform of cytochrome P450
- B) inhibition of the CYP3A isoform of cytochrome P450
- C) genetic polymorphism of an N-acetyltransferase enzyme
- D) competition for Phase I metabolism by phenytoin or phenobarbital
- E) induction of the CYP1A family of P450 isoforms by polycyclic aromatic hydrocarbons

14. Which of the following is NOT CORRECT regarding the hepatotoxicity induced by the analgesic agent acetaminophen (also known as paracetamol)?

- A) can be worsened by inducing cytochrome P450 CYP2E1
- B) can be worsened by fasting
- C) can be attenuated by pretreatment with buthionine-S-sulfoximine or diethyl maleate
- D) is initiated by the covalent binding of a reactive metabolite to mitochondrial proteins
- E) can be fatal in overdose situations in humans if not treated with N-acetylcysteine

15. Which of the following is an INCORRECT match between a toxicant or induced cellular dysfunction and the normal compensatory response?

- A) oxidative stress: inactivation of the protein tyrosine phosphatase (PTEN)
- B) hypoxia: decreased signaling via JNK and p38
- C) DNA damage: activation of p53 protein
- D) energy depletion: activation of AMP-activated protein kinase
- E) electrophiles: release of Nrf2 from Keap1

16. Which one of the following approaches to risk assessment is NOT applicable to genotoxic carcinogens?

- A) linearized multistage model
- B) one-hit linear model
- C) benchmark dose-response modeling with linear extrapolation
- D) NOAEL/LOAEL with uncertainty factors
- E) PBTK modeling

17. The mechanism most likely to be involved in the repair of DNA damage (double-strand breaks) caused by ionizing radiation (such as X-rays, gamma rays, and alpha particles) is which of the following?

- A) base excision repair
- B) nucleotide excision repair
- C) homologous recombination
- D) mismatch repair
- E) O^6 methylguanine-DNA methyltransferase repair

18. A drug substance contains a carcinogenic impurity at 0.1 ppm. If a daily intake 0.15 ug/person/day of this impurity is estimated to associate with a 1×10^{-6} lifetime cancer risk, for a daily dose of the drug at 500 mg/person/day, the lifetime cancer risk for this impurity
- would be greater than 1×10^{-5}
 - would be between 1×10^{-6} and 1×10^{-5}
 - would be between 1×10^{-7} and 1×10^{-6}
 - would be less than 1×10^{-7}
 - cannot be determined based on the information given above
19. The mechanism of gallium arsenide-induced suppression of the humoral immune response involves alteration of the function of which of the following cell types?
- monocytes and polymorphonuclear cells
 - natural killer cells and B cells
 - macrophages, T cells, and basophils
 - macrophages, T cells, and B cells
 - macrophages and natural killer cells
20. Which of the following statements BEST describes the concept of perceived risk?
- rarely incorporated into risk communication decisions
 - a relatively stable personal viewpoint that is not amenable to change
 - unrelated to the precautionary principle
 - highly influenced by factors such as familiarity and controllability
 - can be precisely measured and manipulated
21. In the United States, all Generally Recognized as Safe (GRAS) determinations must be which of the following?
- approved by the Food and Drug Administration
 - kept confidential
 - completed for all color ingredients for which a formal petition will not be sent to the Food and Drug Administration
 - supported by human efficacy data
 - supported by published safety data
22. In the field of epidemiology, the amount of disease that would have never occurred if the exposure had been prevented is measured by which of the following?
- attributable risk
 - proportional mortality ratio
 - relative risk
 - standardized mortality ratio
 - absolute risk
23. According to Good Laboratory Practices (21 CFR 58) reagents must be labeled with which of the following?
- identity, titer or concentration, date opened and expiration date
 - identity, titer or concentration, storage requirements and expiration date
 - identity, titer or concentration, identity of person opening the bottle, and expiration date
 - identity, titer or concentration, identity of person opening the bottle, date opened and expiration date
 - identity, titer or concentration, storage requirements, date opened and expiration date
24. *In vitro* assays for photoreactivity (e.g. reactive oxygen species, ROS) and phototoxicity (e.g. Neutral Red Uptake in BALB/c 3T3 cells) have been characterized as having high sensitivity and low specificity for pharmaceuticals. In predicting potential for phototoxicity associated with pharmaceuticals in humans, which of the following is CORRECT?
- a negative result in either assay is highly predictive of a human phototoxicant
 - a positive result in either assay is highly predictive of a human phototoxicant
 - a negative photoreactivity result combined with a positive *in vitro* phototoxicity result is highly predictive that the pharmaceutical is not a phototoxicant in humans
 - a negative result in an *in vitro* phototoxicity test is highly predictive that the pharmaceutical is not a phototoxicant in humans
 - absorbance of UVB, UVA, or visible radiation light (290-700 nm) is highly predictive of a human phototoxicant
25. Which of the following is CORRECT regarding quantitative structure-activity relationship (QSAR) evaluations?
- QSAR analysis is reliable when predictions are made for chemicals outside of the domain of applicability
 - QSAR models are capable of making predictions for a wide range of chemicals
 - predicting endpoints with diverse mechanisms is less complex than predicting endpoints with limited mechanisms
 - predicting endpoints with limited mechanisms is less complex than predicting endpoints with diverse mechanisms
 - QSAR models accurately predict toxicity of metallic compounds

26. In a 2-year inhalation study, rats and mice were exposed to Chemical X for 8 hours/day, 5 days/week. The study found a No Observed Adverse Effect Level (NOAEL) of 300 ppm. In order to protect the general worker population, a permissible exposure limit (PEL) of 3.00 ppm for an 8 hour day was established by the Occupational Safety and Health Administration. What is the appropriate exposure limit for off-shore oil rig workers working 12 hour shifts?

- A) 3.00 ppm
- B) 2.40 ppm
- C) 0.30 ppm
- D) 2.00 ppm
- E) 2.25 ppm

27. The objective of a toxicology study is to determine whether a treatment elicits a response. Which of the following statements regarding toxicological data generation and statistical analysis of those data is CORRECT?

- A) an observed difference in response between treated and control is necessarily the result of treatment
- B) an observed difference in response between treated and control is either the result of treatment or experimental bias
- C) a major objective of both experimental design and statistical analysis is to minimize effects of chance
- D) the smaller the probability of a type II error, the more confident the researcher can be that the effect is real
- E) the smaller the probability of a type I error, the more confident the researcher can be that the effect is real

28. Sulfur dioxide is considered a component of outdoor ambient air pollution. Which of the following statements below best characterizes the toxicological properties of this substance at environmentally-relevant concentrations?

- A) sulfur dioxide is a water-soluble irritant gas absorbed predominantly in the upper airways and can stimulate bronchoconstriction
- B) sulfur dioxide is an insoluble particulate absorbed predominantly in the upper airways and can stimulate bronchoconstriction
- C) sulfur dioxide is a non-irritating water-soluble gas that is not absorbed in the airways or dermally to any significant extent and stimulates bronchoconstriction only in individuals with existing asthmatic or emphysematous conditions
- D) sulfur dioxide is an insoluble particulate absorbed predominantly by swallowing particles trapped in the respiratory mucous of the upper airways and typically causes erosion of the gastric mucosa
- E) sulfur dioxide in the atmosphere is primarily due to production by cement manufacturing and smelting operations

29. Experimental fish screening assays for endocrine active substances are most likely to utilize which of the following specific indicators to demonstrate exogenous estrogen simulation effects?

- A) synthesis of vitellogenin
- B) plasma cholinesterase inhibition
- C) induction of mixed function oxygenases
- D) delta-aminolevulinic acid dehydratase inhibition
- E) induction of metallothionein

30. Which of the following environmental contaminants has been associated with androgenic activity?

- A) methoxychlor
- B) phthalates
- C) pulp/paper mill effluents (PME)
- D) PCBs
- E) p,p'-DDE

31. Which of the following is CORRECT about the gastrointestinal (GI) absorption of substances?

- A) drugs, chemicals, minerals, etc. are absorbed from the GI primarily by active transport with expenditure of ATP
- B) P-glycoprotein (P-gp) and multi-drug resistant protein 2 (MRP2) are efflux GI transporters
- C) the ATP-binding cassette (ABC) transporters are primarily influx transporters of substances
- D) thallium transport competes with the calcium transporter for cellular uptake
- E) the flavinoid naringin in grapefruit juice induces GI CYP3A leading to decreased drug absorption

32. Which of the following is NOT CORRECT regarding microRNAs (miRNAs)?

- A) urinary miRNAs have been shown to correlate with liver injury induced by acetaminophen (APAP) or carbon tetrachloride (CCl₄) in rats
- B) miRNAs are approximately 21-25 nucleotides in length, single-stranded RNA that regulate gene expression by targeting messenger RNAs (mRNAs) by binding complementary regions of their transcript
- C) miRNA has two primary regions to include the 5' region 'seed sequence' and the 3' 'skewed region'
- D) one of the first studies to identify that miRNAs regulate CYP enzymes was reported for human CYP1B1
- E) recently, several studies have identified that miRNAs play a critical role in the differentiation and function of the adaptive and innate immune system

33. As a highly reactive substance, inhaled formaldehyde directly targets diverse nucleophiles of DNA, forming DNA adducts, a key event initiating cancer. Determining the internal dose of formaldehyde received by exposed humans is important for both risk assessment and biomarker development purposes. However, quantifying the number of DNA adducts caused by inhalational exposure to formaldehyde has proven difficult due to which of the following?

- A) the methods for isolating DNA adducts from cells are still not sufficiently optimized, producing unreliable data that cannot be reproduced between laboratories
- B) a substantial number of endogenously-formed formaldehyde adducts is present in the body due to the role of formaldehyde as an essential metabolic intermediate in living cells
- C) reactive oxygen species are continuously formed in all cells from metabolic and other biochemical reactions, confounding quantification of DNA adducts
- D) the detection limits of chromatography and mass spectrometry instruments are not sensitive enough to accurately quantify DNA adducts
- E) estrogen metabolites can cause DNA damage in a way that is similar to formaldehyde-induced DNA damage

34. Of the following, which is considered to be the most definitive diagnostic tool for assessing the potential for a chemical to cause autoimmune disease?

- A) number of CD4⁺ lymphocytes
- B) the popliteal lymph node assay (PLN)
- C) immunohistopathology
- D) the local lymph node assay (LLNA)
- E) human lymphocyte transformation assay

35. Which of the following is CORRECT regarding OSHA Permissible Exposure Limits?

- A) are updated on a regular basis
- B) exist for the vast majority of airborne chemicals found in US workplaces
- C) are set as 10-hour time-weighted-average concentrations
- D) are legally enforceable standards in the US
- E) have been in use in the US since the 1940s

36. Which of the following toxicology studies is no longer required prior to filing a successful Investigational New Drug (IND) application for a Phase I clinical trial in healthy volunteers?

- A) repeat-dose toxicity study in two species (one rodent and one nonrodent)
- B) cardiovascular safety pharmacology
- C) Ames reverse mutation assay
- D) single-dose toxicity studies by oral and/or intravenous routes in two rodent species
- E) central nervous system safety pharmacology

37. When determining the highest dose to be evaluated in acute, sub-chronic and chronic toxicity studies for pharmaceuticals, which of the following statements best represents the highest dose that should be selected according to ICH guidance?

- A) it is necessary to demonstrate the MTD (maximum tolerated dose) in every study
- B) it is necessary to demonstrate a safety margin of 100 fold beyond the maximum clinical exposure
- C) the highest dose in most cases should be limited to 1000 mg/kg/day
- D) it is necessary to demonstrate a safety margin of 1000 fold beyond the maximum clinical exposure
- E) it is necessary to demonstrate the MFD (maximum feasible dose) in every study

38. The United States Animal Welfare Act and Animal Welfare Regulations (Code of Federal Regulations, Title 9, Chapter 1, Subchapter A) lists which of the following as a requirement for the Institutional Animal Care and Use Committee (IACUC)?

- A) the Chairman of the IACUC appoints IACUC members
- B) the IACUC must inspect annually all of the research facility's animal facilities
- C) at least one member of the IACUC must not be affiliated with the facility
- D) the IACUC must be comprised of a Chairman and at least three additional members
- E) IACUC members must serve at least three years of appointment

39. Which of the following associations (toxic syndromes -classic clinical features) is NOT correct?

- A) narcotic - mydriasis, hyperalert, no change in pulse, no change in blood pressure
- B) anticholinergic - mydriasis, altered mental status, increased pulse, no change in blood pressure
- C) sympathomimetic - mydriasis, hyperalert, increased pulse, increased blood pressure
- D) opioid - miosis, decreased level of consciousness, decreased pulse, no change in blood pressure
- E) cholinergic - miosis, decreased pulse, altered mental status

40. Which of the following is NOT CORRECT regarding ethanol in forensic toxicology laboratories?

- A) ethanol can be quantitated in urine by osmometry
- B) breath ethanol concentrations reflect blood concentrations and can be measured from alveolar air (end-expired air) by infrared techniques
- C) a blood concentration of 500 mg ethanol per 100 ml of blood is considered a probable fatal level
- D) women have lower levels of alcohol dehydrogenase in the gastric mucosa
- E) false positive levels of ethanol in urine may be attributed to *Escherichia coli* contamination in this sample